

Full Solution of Two General-Compact Converse Problems for Weak-Star-to-Weak Points of Continuity

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Status and source questions

Status: full solution, likely valid, pending human review.

Let Ω be nonempty compact Hausdorff and X a nonzero Banach space. Write $\text{WC}(\Omega, X)$ for the Banach space of bounded weakly continuous maps $\Omega \rightarrow X$, with the supremum norm. A functional $x^* \in B_{E^*}$ is a *weak-star-to-weak point of continuity* (abbreviated w^* - w PC) when the identity

$$(B_{E^*}, w^*) \longrightarrow (B_{E^*}, w)$$

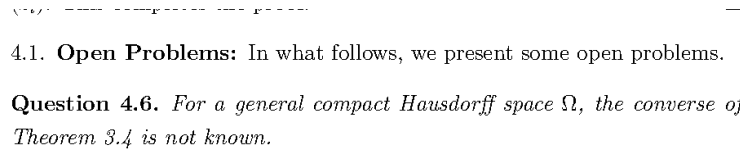
is continuous at x^* .

The source works under the standing convention that Ω is infinite. Theorems 3.4 and 4.1 of Dwivedi [1] prove necessary atomic descriptions of the w^* - w PCs in the dual balls of $C(\Omega, X)$ and $\text{WC}(\Omega, X)$, respectively. On PDF pages 15–16 the paper asks:

Question 4.6. For a general compact Hausdorff space Ω , is the converse of Theorem 3.4 true?

Question 4.7. For a general compact Hausdorff space Ω , is the converse of Theorem 4.1 true?

The source establishes both converses under extremal disconnectedness. We prove that this topological assumption is unnecessary.



4.1. **Open Problems:** In what follows, we present some open problems.

Question 4.6. For a general compact Hausdorff space Ω , the converse of Theorem 3.4 is not known.

Figure 1: Question 4.6 on source PDF page 15.

Question 4.7. *For a general compact Hausdorff space Ω , the converse of Theorem 4.1 is not known.*

Figure 2: Question 4.7 on source PDF page 16.

Proof intuition

All relevant functionals live on isolated points. If I is the set of isolated points of Ω , then zero extension identifies $c_0(I, X)$ with a subspace of both ambient function spaces. A pointwise three-ball argument shows that this subspace is always an M -ideal; density of I and clopenness of $\bar{I}$ are not needed.

On the dual $\ell_1(I, X^*)$, coordinatewise w^* convergence by itself is not enough for weak convergence. The decisive extra fact is that the coordinate norms of the proposed limit add up to the entire unit norm. Lower semicontinuity then forces convergence of each coordinate norm and uniform smallness of the tails. The coordinate PC hypothesis handles every finite head, and the tail bound finishes the weak-convergence proof. The standard M -ideal lifting mechanism then transfers the PC to either ambient dual. For completeness, the finite-space edge case is separated in the final theorem: when both Ω is finite and X is reflexive, the ambient space is reflexive and every point of its dual ball is a PC.

The isolated-coordinate M -ideal

We use the standard three-ball characterization of M -ideals [3, Chapter I].

Lemma 1 (isolated-coordinate lemma). *Let S be a set, let $I \subseteq S$, and let E be a closed linear subspace of $\ell_\infty(S, X)$ which contains the zero extensions of $c_0(I, X)$. Then that copy of $c_0(I, X)$ is an M -ideal in E .*

Proof. Put $Y = c_0(I, X)$, extended by zero on $S \setminus I$. Take $z \in B_E$, $y_1, y_2, y_3 \in B_Y$, and $\varepsilon > 0$. There is a finite $F \subseteq I$ such that

$$\|y_j(i)\| < \varepsilon \quad (i \in I \setminus F, 1 \leq j \leq 3).$$

Define $y \in Y$ by $y(i) = z(i)$ for $i \in F$ and $y(s) = 0$ otherwise. For $s \in F$,

$$(z + y_j - y)(s) = y_j(s),$$

so its norm is at most one. For $s \in I \setminus F$ its norm is at most $1 + \varepsilon$, and for $s \in S \setminus I$ it equals $\|z(s)\|$. Consequently

$$\|z + y_j - y\| \leq 1 + \varepsilon \quad (1 \leq j \leq 3).$$

This is the three-ball property, hence Y is an M -ideal in E . □

If I is the set of isolated points of a compact Hausdorff space Ω , zero extension maps $c_0(I, X)$ into $C(\Omega, X)$. Indeed, for $u \in c_0(I, X)$ and $\varepsilon > 0$, the set $\{i \in I : \|u(i)\| \geq \varepsilon\}$ is finite and clopen in Ω ; this proves continuity at every point outside I . In particular, the zero extensions also belong to $WC(\Omega, X)$. Lemma 1 therefore applies to both ambient spaces.

Atomic PCs in the ℓ_1 -sum

Lemma 2 (full-mass atomic PC). *Let $J \subseteq I$ be nonempty, finite or countable. Let $(x_i^*)_{i \in J} \subseteq S(X^*)$ consist of w^* -w PCs of B_{X^*} , and let $\alpha_i > 0$ with $\sum_{i \in J} \alpha_i = 1$. Regard*

$$\lambda = (\alpha_i x_i^*)_{i \in J}$$

as an element of $B_{\ell_1(I, X^)}$ by putting zero in the other coordinates. Then λ is a w^* -w PC of $B_{c_0(I, X)^*} = B_{\ell_1(I, X^*)}$.*

Proof. Let $u_\gamma = (u_{\gamma, i}^*)_{i \in I}$ be a net in $B_{\ell_1(I, X^*)}$ converging w^* to λ . For every $i \in J$, testing against the i th coordinate copy of X gives

$$u_{\gamma, i}^* \xrightarrow{w^*} \alpha_i x_i^*.$$

For each finite $F \subseteq J$, weak-star lower semicontinuity of the norm gives

$$\liminf_{\gamma} \sum_{i \in F} \|u_{\gamma, i}^*\| \geq \sum_{i \in F} \alpha_i. \quad (1)$$

Fix $i \in J$. The one-coordinate case of (1) gives the lower bound for $\|u_{\gamma, i}^*\|$. If G is any finite subset of $J \setminus \{i\}$, then

$$\|u_{\gamma, i}^*\| \leq 1 - \sum_{j \in G} \|u_{\gamma, j}^*\|.$$

Taking the upper limit and then finite G whose α -mass tends to $1 - \alpha_i$ yields

$$\|u_{\gamma, i}^*\| \longrightarrow \alpha_i.$$

Thus, eventually normalizing the i th coordinate gives

$$\frac{u_{\gamma, i}^*}{\|u_{\gamma, i}^*\|} \xrightarrow{w^*} x_i^*.$$

The PC hypothesis on x_i^* upgrades this to weak convergence, so $u_{\gamma, i}^* \rightarrow \alpha_i x_i^*$ weakly for every $i \in J$.

It remains to make the convergence uniform over the coordinates. Given $\eta > 0$, choose finite $F \subseteq J$ with $\sum_{i \in F} \alpha_i > 1 - \eta$. Equation (1) implies that, eventually,

$$\sum_{i \notin F} \|u_{\gamma, i}^*\| < 2\eta, \quad \sum_{i \notin F} \alpha_i < \eta.$$

Now take any $(x_i^{**})_{i \in I} \in \ell_\infty(I, X^{**})$. The sum over F converges by coordinatewise weak convergence, while the two tails have absolute value at most $3\eta \sup_i \|x_i^{**}\|$. Since η is arbitrary, $u_\gamma \rightarrow \lambda$ weakly in $\ell_1(I, X^*)$. $\square$

Lifting through an M -ideal

For completeness we record the short lifting step, also appearing as Lemma 3.1 of the source via [2, Theorem 6.2].

Lemma 3 (PC lifting). *If Y is an M -ideal in E , then a w^* - w PC $\lambda \in S(Y^*)$, viewed in the canonical L -summand of E^* , is a w^* - w PC of B_{E^*} .*

Proof. Write $E^* = Y^\perp \oplus_1 Y^\#$, where $Y^\#$ is the canonical copy of Y^* . Suppose $a_\gamma + b_\gamma \in B_{E^*}$ converges w^* to $\lambda \in Y^\#$. Restriction to Y shows that $b_\gamma \rightarrow \lambda$ w^* in B_{Y^*} , hence weakly by the PC property. Moreover,

$$1 = \|\lambda\| \leq \liminf_\gamma \|b_\gamma\|, \quad \|a_\gamma\| + \|b_\gamma\| \leq 1,$$

so $\|a_\gamma\| \rightarrow 0$. It follows that $a_\gamma + b_\gamma \rightarrow \lambda$ weakly in E^* . $\square$

Full classification

Theorem 4. *Let Ω be a nonempty compact Hausdorff space, let X be a nonzero Banach space, and let E be either $C(\Omega, X)$ or $WC(\Omega, X)$.*

If Ω is finite and X is reflexive, every point of B_{E^} is a w^* - w PC. In every other case, a functional $\Lambda \in B_{E^*}$ is a w^* - w PC if and only if it has the canonical form, for a nonempty finite or countable index set J ,*

$$\Lambda = \sum_{n \in J} \alpha_n \delta_{\omega_n} \otimes x_n^*, \tag{2}$$

where the ω_n are pairwise distinct isolated points of Ω , $\alpha_n > 0$, $\sum_n \alpha_n = 1$, and every $x_n^ \in S(X^*)$ is a w^* - w PC of B_{X^*} .*

Proof. If Ω is finite and X is reflexive, then $E = \ell_\infty(\Omega, X)$ is reflexive, so the weak-star and weak topologies coincide on all of B_{E^*} .

We next prove necessity outside this exceptional case. For infinite Ω , it is Theorem 3.4 of [1] for $E = C(\Omega, X)$ and Theorem 4.1 there for $E = WC(\Omega, X)$. Suppose instead that Ω is finite and X is nonreflexive. Then $E = \ell_\infty(\Omega, X)$ is nonreflexive. A w^* - w PC $u^* \in B_{E^*}$ must have norm one: otherwise, using the norm slack at u^* , continuity there would make every bounded weak-star-null net weakly null after scaling. It would follow that weak-star convergence implies weak convergence on B_{E^*} , making B_{E^*} weakly compact and E reflexive, a contradiction. Write $u^* = (u_\omega^*)_{\omega \in \Omega} \in \ell_1(\Omega, X^*)$. If $u_\omega^* \neq 0$, replace its coordinate by $\|u_\omega^*\|v_\gamma^*$ for an arbitrary net $v_\gamma^* \in B_{X^*}$ converging weak-star to $u_\omega^*/\|u_\omega^*\|$. The resulting net remains in B_{E^*} and converges weak-star to u^* ; the PC property and the coordinate projection therefore show that v_γ^* converges weakly. Thus every normalized nonzero coordinate is a PC, which is exactly the stated canonical form.

For sufficiency, let I be the set of isolated points and let $Y = c_0(I, X)$, extended by zero. Lemma 1 says that Y is an M -ideal in either choice of E . Under $Y^* = \ell_1(I, X^*)$, the restriction of (2) is exactly the full-mass atomic functional of Lemma 2; hence it is a w^* - w PC of B_{Y^*} . Its canonical extension in the L -summand of E^* is precisely (2). Lemma 3 completes the proof. $\square$

Thus both Questions 4.6 and 4.7 have affirmative answers. In particular, the source's topological hypothesis is unnecessary. The extra finite-space clause above supplies the literal all-compact classification beyond the source's standing infinite-space convention.

Verification and limitations

The proof uses nets throughout. It allows uncountable I : only the countable support of (2) enters the finite-head argument, while the tail estimate controls every coordinate outside a chosen finite subset. Pairwise distinct support points are part of the canonical support decomposition. If a display repeats a point, its atoms must first be combined, and the normalized combined coordinate must satisfy the PC hypothesis.

The three-ball estimate was checked in its three pointwise regimes: on the finite copied set, on the remaining isolated points, and off the isolated set. The finite/reflexive exception was checked separately rather than being hidden inside the source’s infinite-space convention. No computational verification is relevant.

This packet does not address Questions 4.8 or 4.9 of the source.

Bounded novelty search

On 9 August 2026 we searched the run’s registry, solution, attempt, and proof-gap indexes for arXiv:2604.09133 and the core terms. Web searches used the exact paper title, the quoted converse formulations, w^* - w PC terms for $C(K, X)$ and $WC(K, X)$, and the $c_0(I, X)/M$ -ideal mechanism. They found the source preprint and older background literature, including [2], but no separate paper explicitly answering either general converse. This supports, but does not guarantee, novelty.

Human-review focus

The three decisive checks are: (i) applying the standard three-ball characterization in Lemma 1; (ii) the norm-saturation and tail argument in Lemma 2; and (iii) identifying the atomic functional with the canonical L -summand extension in Lemma 3.

References

- [1] S. Dwivedi, *An extension of Phelps theorem to spaces of vector-valued functions*, arXiv:2604.09133v1 (2026), <https://arxiv.org/abs/2604.09133>.
- [2] S. Daptari, V. Montesinos, and T. S. S. R. K. Rao, *A study of weak*-weak points of continuity in the unit ball of dual spaces*, Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. 119 (2025), 82; arXiv:2506.08458, <https://arxiv.org/abs/2506.08458>.
- [3] P. Harmand, D. Werner, and W. Werner, *M-Ideals in Banach Spaces and Banach Algebras*, Lecture Notes in Mathematics 1547, Springer, 1993.